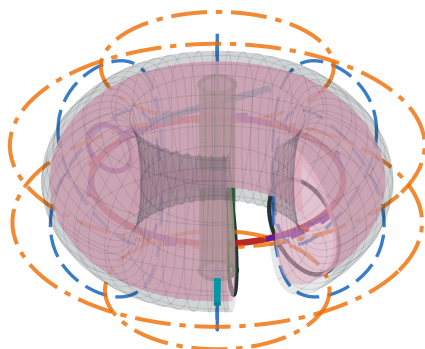


High-field design state — the inboard coil acquires a technology requirement

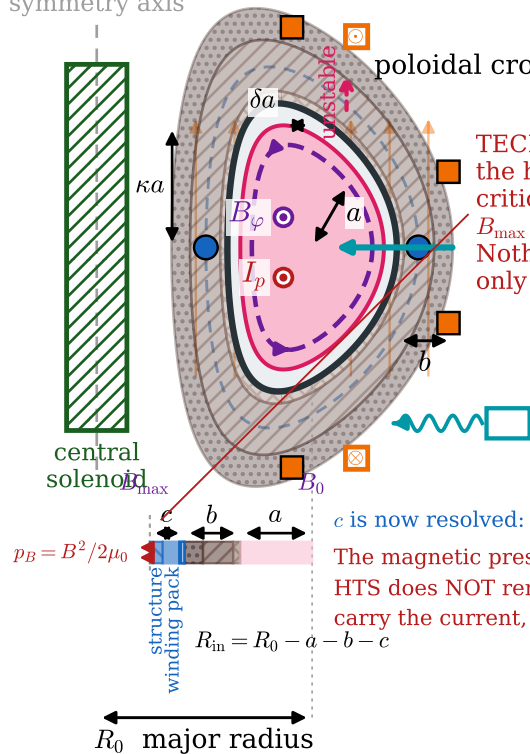
the same machine as stage 14. Only the coil bracket c is resolved further, and the demand it now carries is named

perspective view (vessel cut open)



- vacuum vessel (chamber wall)
- plasma / fuel region
- toroidal-field (TF) coils
- central solenoid (CS)
- poloidal-field (PF) coils
- magnetic field $\mathbf{B}$
- plasma current I_p
- auxiliary heating ports

symmetry axis



poloidal cross-section (R - Z plane)

TECHNOLOGY REQUIREMENT, not an achieved property:
the high-field branch needs a conductor whose critical-current margin still supports $B_{\max} \gtrsim 17.6$ T at the winding pack.
Nothing in this figure supplies it. The figure only says where the demand lands.

c is now resolved: structural case | winding pack | insulation and cooling

The magnetic pressure $p_B = B^2/2\mu_0$ still presses on the case.

HTS does NOT remove that force: it changes what conductor can carry the current, not what the structure has to hold.

Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.